

Evaluating LLMs as annotators

Deep Learning in NLP
(Groups 2 and 3, RISS annotation scheme)

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Outline

1. Preliminaries
 - setting the task
 - interim annotation output description
2. Post-processing
 - annotations → dataset
 - IAA methodology
 - IAA results (trial session): per-category and per-tag definitions
 - Label distribution analysis
3. Evaluating LLM methods and models
 - Prompting
 - Finetuning
4. Project links and definitions
5. Examples and trivia

Workflow overview

You are here

Building a corpus

1. text+meta acquisition
2. sampling
3. coding scheme design
4. human annotation
5. postprocessing, IAA
6. gold standard



LLMs evaluation

1. method and model selection
2. comparison against human performance



Best LLM application to the full corpus

1. scaling human annotation
2. building a dataset



Social-scientific analysis

using the dataset to answer main research question

Main Groups2-3: Human annotation throughput

Current annotation statistics for Hansard (1980, 2025, 2026)

Parameter	Value
Available items	520
Annotated items	71
Triple+ annotated	3 (4.23%)
Double-annotated	27 (38.03%)
Annotations	104
Annotators	4
Comments	1
Avg. min/item	3.47
Project start	2026-06-16
Project end	2026-06-18 10:15

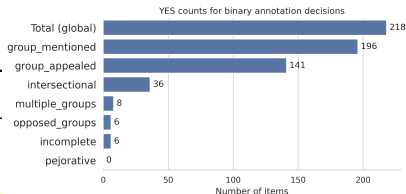
aka `main_student_groups`

IAA does not make sense on < 30-50 cross-annotated items.

UPD Main Groups2-3: Human annotation throughput

Updated annotation statistics for Hansard (1980, 2025, 2026)

Parameter	Value
Available items	520
Annotated items	218
Triple+ annotated	124 (56.88 %)
Double-annotated	74 (33.94%)
Annotations	540
Annotators	7
Comments	4
Avg. min/item	1.32
Project start	2026-06-16
Project end	2026-06-23 01:06



	Alias	Items
group2	Horse	10
	Tiger	0
	Zebra	119
	Whale	37
group3	Robin	181
	Crane	0
	Raven	181
	lecturers	12

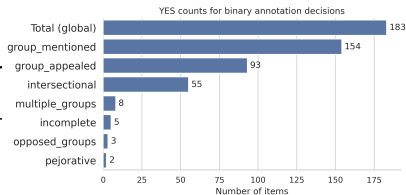
aka **main_student_groups**
gold standard: **98 items**

Trial Session: Human annotation throughput

First attempt annotation statistics for Hansard (1980, 2025, 2026)

Parameter	Value
Available items	60x2+120x2
Annotated items	183
Annotations	483
Annotators	10
Comments	33
Triple+ annotated	67 (36.6%)
Double-annotated	7 (3.8%)
Avg. time/item	3.24 min
Project period	2–11 Jun 2026

aka **trial**_student_groups



	Alias	Items
group2	Horse	60
	Tiger	60
	Zebra	60
	Whale	6
group3	Robin	180
	Crane	45
	Raven	20
	lecturers	52

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Formalising human-facing annotation setup

Annotation interface:

Task 1: Is a social group invoked as part of a political argument?

Task 2: Which social group is invoked?

Task 3: How is the group invoked?



Suggested dataset structure: 10 decisions/fields
(6 binary variables + 1 multi-label variable + 3 multiclass variable)

yes-no outcomes:

1. group mentioned?
2. group appealed?
3. intersectional?
4. multiple groups?
5. opposed groups?
6. pejorative reference?

multi-label/class outcomes:

7. group type (multi-label)?
8. appeal type: reasoning?
9. appeal type: polarity?
10. appeal type: stance?

Each annotated item (==sentence ID) is a row in a table
with 10 annotated properties + context + meta fields

	column	value
1.	sent_id	lords_2025_13095:004
2.	text	<i>I welcome the provisions in the Bill setting out child protection and safeguarding. . .</i>
...	...	...
10.	group_mentioned	yes
11.	group_appealed	yes
12.	group_tags	[children]
13.	reasoning	cultural
13.	polarity	pro-group
13.	stance	endorse
14.	intersectional	no
15.	multiple_groups	no
16.	opposed_groups	no
17.	pejorative	no

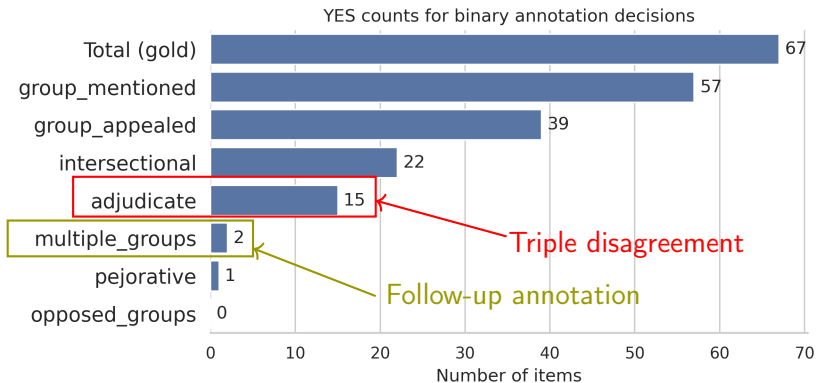
NB! Use internal tags; interface labels are not guaranteed to be unique.

Cross-annotations and adjudication

Every sent_id:

3 annotations → gold standard via adjudication

- Binary and multiclass variables: majority vote
- Tag lists: majority vote per tag
 - Exact-list match is too strict; tag union too permissive.
- No majority: adjudicate=True



IAA: Inter-Annotator Agreement

Annotation quality considerations

- **Validity:** Are we capturing the intended concept? A question of sociological theory knowledge and of ultimate RISS research questions.
- **Reliability:** Do annotators apply the scheme consistently?
- **Reproducibility:** Can similar results be obtained again with a different sample, team, or run?

Inter-Annotator Agreement (IAA)

– is reliability measure, not validity. Three annotators can agree perfectly on an incorrect interpretation.

High IAA $\Rightarrow$ annotation scheme is applied consistently

Low IAA $\Rightarrow$ ambiguity in data (legit interpretability), overlapping categories, or unclear guidelines

IAA metrics and interpretation

Percentage agreement

- Proportion of items receiving identical labels.
- Easy to interpret, but ignores agreement by chance.

Cohen's κ

- Corrects observed agreement for chance agreement.
- Suitable for two annotators and complete annotation.
- Sensitive to class imbalance and prevalence effects.

Krippendorff's α (main metric)

- Chance-corrected reliability coefficient.
- Handles > 2 annotators, missing values, and diverse scales.
- Widely recommended for content analysis.

Rule-of-thumb interpretation (Krippendorff, 2004³; Artstein, 2008¹)

α value	Interpretation
≥ 0.80	Reliable agreement
0.67–0.79	Tentatively acceptable
< 0.67	Insufficient reliability

Krippendorff's α (nominal): formal definition

α compares **observed disagreement** (D_o) with **disagreement expected by chance** (D_e).

$$\alpha = 1 - \frac{D_o}{D_e}$$

$$D_o = \frac{\sum_u \sum_c n_{uc}(n_u - n_{uc})}{\sum_u n_u(n_u - 1)}$$

$$D_e = \frac{\sum_c n_c(N - n_c)}{N(N - 1)}$$

- D_o : observed proportion of disagreeing coder pairs
- D_e : baseline disagreement implied by the label distribution
- u indexes items; c indexes categories.
- n_{uc} : annotators assigning category c to item u .
- n_u : valid annotations for item u .
- n_c : assignments to category c across all items.
- $N = \sum_c n_c$: total number of valid annotations.

Types of annotation variables

Type	Examples	Values per item
Binary	group_mentioned, group_appealed	0 or 1 value
Multiclass	reasoning, polarity, stance	exactly 1 value
Multi-label	group_tags	0, 1, or many values

Examples

- Binary: group_mentioned = yes
- Multiclass: reasoning = REASONING_ECONOMIC
- Multi-label: group_tags = [AGE_CHILD, INC_LOW]

Implications for agreement analysis

- Binary and multiclass variables are evaluated directly.
- Multi-label variables can be evaluated:
 - at the category level (e.g. age, family, migration);
 - at the individual tag level (e.g. AGE_CHILD).

This is key for: IAA, majority voting, disagreement score, LLM evaluation

α validity conditions

- **Usable item:** an item (sent_id) with at least two valid annotations.
- **Usable value:** a category value observed among usable items.
- α is not computed if there are no usable items
 - status: **insufficient_overlap**
- α is not computed if only one usable value is observed
 - status: **single_value**
- Category-level α is not computed when fewer than two category values occurred at least twice
 - status: **too_sparse**
- Category-level α based on very few usable items should be interpreted with caution.
 - e.g. *family*: $\alpha = 1.0$ based on only 3 usable items
- Per-tag α is not computed for tags with **frequency** < 5 .
 - space considerations + no unstable estimates for rare labels

Trial Session: IAA for binary and multi-class categories

Appeal types: 3-4 values in each category are mutually exclusive

(in the Label Studio interface order)

type	variable	agree_pct	α
binary	group_mentioned	71.64	0.346
binary	group_appealed	49.25	0.074
binary	intersectional	53.73	0.340
binary	multiple_groups	89.55	0.093
binary	opposed_groups	97.01	-0.005
binary	pejorative	98.51	0.497
multiclass	reasoning	60.00	0.467
multiclass	polarity	75.61	0.420
multiclass	stance	53.85	0.086

($N = 67$ triple-annotated items)

Category-level IAA for a multi-label category

Social group type is a **multi-label category**:

- items may receive multiple social-group category labels;
- for category-level IAA, each category is evaluated separately as a nominal variable, collapsing individual labels

category	agree_pct	α	usable items	usable values	status
age	57.14	0.529	7	5	ok
gender	100.00	1.000	5	2	ok
family	100.00	1.000	3	3	ok
disability	80.00	0.629	5	3	ok
ethnicity	NaN	NaN	2	1	single_value
migration	66.67	0.533	3	2	ok
religion	NaN	NaN	4	1	single_value
geography	NaN	NaN	0	0	insufficient_overlap
education	NaN	NaN	3	1	single_value
income	66.67	0.489	6	3	ok
occupation	66.67	0.655	6	8	ok

Trial Session: Per-tag IAA on social group types

How often do annotators agree on the presence of each group tag?

Krippendorff's α for social-group labels (frequency ≥ 5)

category	variable	agree_pct	α	tag_freq
age	school age	91.04	0.223	8
age	children	91.04	0.306	9
gender_sexuality	women	98.51	0.904	11
family	families	92.54	0.352	8
disability	disabled people	97.01	0.824	12
ethnicity	ethnic minority	91.04	0.372	10
migration	refugees	98.51	0.796	5
religion	jewish	97.01	0.791	10
education	in education	95.52	0.611	8
income	poor/low-income	97.01	0.592	5
income	precarious/unemployed	92.54	0.352	8
occupation	agriculture	98.51	0.853	7
other_groups	non-RISS	34.33	0.108	85

Tag coverage for group and appeal variables I

in triple-annotated data

	seen	unseen	total
Coverage: 65.5%			
group type	44	29	73
appeal type	11	0	11
overall	55	29	84

in global annotations

	seen	unseen	total
Coverage: 71.4%			
group type	49	24	73
appeal type	11	0	11
overall	60	24	84

Tag coverage for social group category II

Tags never seen globally in any of 483 annotations

category	total	seen	unseen	unseen (%)
age	8	7	1	12.5
gender_sexuality	7	3	4	57.1
family	9	3	6	66.7
disability	3	3	0	0.0
ethnicity	2	2	0	0.0
migration	8	4	4	50.0
religion	5	4	1	20.0
geography	5	3	2	40.0
education	6	4	2	33.3
income	6	6	0	0.0
occupation	13	9	4	30.8
other_groups	1	1	0	0.0

Tag frequency for group and appeal variables III

Seen but Rare (based on triple-annotated subset)

α is not computed for tags with frequency < 5 or zero variance

Proportion of seen group labels excluded from IAA analysis

category	low freq./total	(%)
age	4/6	66.7
gender_sexuality	1/2	50.0
family	2/3	66.7
disability	2/3	66.7
ethnicity	1/2	50.0
migration	3/4	75.0
religion	2/3	66.7
geography	3/3	100.0
education	2/3	66.7
income	3/5	60.0
occupation	8/9	88.9
other_groups	0/1	0.0

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Experimental design

Open-weight models (instruction-tuned)

- Qwen3-1.7B⁶
- Qwen3.6-27B⁵
- gemma-4-31b-it²
- gpt-oss-120b⁴

Methods

- Zero-shot (using short definitions)
- Few-shot (using examples)
- Fine-tuning (not sure it is feasible)

Models $\times$ Methods (on valid gold items)



LLM Annotation Quality based on LLM-to-gold IAA

Available gold data: 52 auto-adjudicated sent_ids

Trial Session: 67 triple-annotated - 0 incomplete - 15 no-majority = 52

LLM annotation task definition

Goal: Use an instruction-tuned LLM as an automatic annotator and compare its predictions against the human gold standard.

Task decomposition

1. Binary classification

- group mentioned?
- group appealed to?
- intersectional appeal?
- multiple groups?
- opposed groups?
- pejorative reference?

2. Multi-label classification

- assign all applicable social-group tags

3. Multiclass classification

- reasoning: economic / cultural / both / unclear
- polarity: pro-group / anti-group / mixed
- stance: endorse / reject / partial / unclear

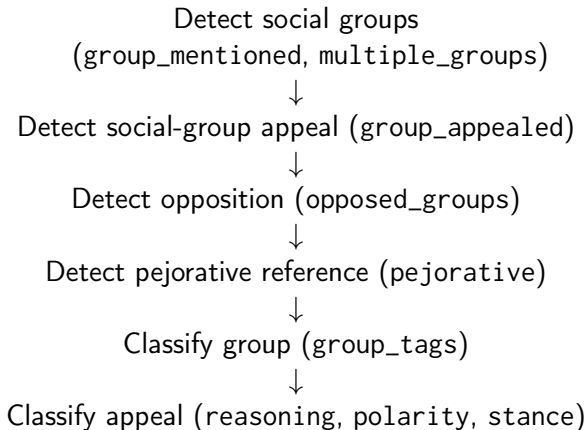
This decomposition mirrors the human annotation workflow and allows separate evaluation of different annotation decisions.

Expected result: default formulation

```
annotations = {  
    "groups": [],  
    "group_mentioned": "no",  
    "multiple_groups": "no",  
    "first_group": None,  
    "group_appealed": "no",  
    "opposed_groups": "no",  
    "pejorative": "no",  
    "group_tags": [],  
    "intersectional": None,  
    "reasoning": None,  
    "polarity": None,  
    "stance": None,  
}
```

Hierarchical annotation pipeline

Take a step up from what you may or may not have implemented in May for 5-class emotion classification.



Pipeline terminates if no group mention or no appeal is detected.

Some detected failure modes for mentions extraction

This fails for single-mention sentences, but works for multiple:
`{"group": []}`

List social groups mentioned in the TARGET sentence, **even if there is one group.**

Output

Ex.1: "groups": ["disabled people"]

Ex.2: "groups": ["agricultural and food industries", "farmers", "workers"]

You cannot currently connect to a GPU due to usage limits in Colab. (Colab does not publish these limits, in part because they can vary over time.)

Omitting the rest for brevity

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Omitting the rest for brevity

Take-aways from zero-shot prompting

1. The model can recognise social groups in simple formulations.
 - ↪ Start with the simplest prompt that solves the task.
2. Small prompt changes can produce qualitatively different outputs. + Output format matters much more than expected.
 - ↪ Validate prompts on a small set of carefully selected examples.
3. A mega annotation prompt might be less reliable than a staged pipeline, esp. for decision hierarchy.
 - ↪ a trade-off between controlled prompting and tokens budget!
4. JSON schemas that look equivalent to humans are not equivalent to the model.
 - ↪ avoid abstract and wordy prompts, don't copypaste LabelStudio instructions.

Few-shot prompting

- Split the gold data into training and test sets.
- Use a stratified sampling strategy rather than a purely random split.
- Select representative examples from the training set.
- Augment each step of the zero-shot annotation pipeline with a small number of annotated examples.
- Evaluate performance on the held-out test set.

Example incorporated into the prompt

TARGET: Disabled people will receive additional support.

Output:

```
{"groups": ["disabled people"]}
```

TARGET: Farmers and workers face rising costs.

Output:

```
{"groups": ["farmers", "workers"]}
```

NB: The methodology and prototypes are developed. Large-scale experimental runs were limited by available Colab runtime and compute resources.

Fine-tuning as an annotation strategy

Human annotations → Instruction–response examples → Supervised fine-tuning → Automatic annotator

Training workflow

1. Use the previous train-test split of the gold dataset.
2. Convert each annotated item into an instruction–response pair.
3. Fine-tune the model on the training data.
4. Apply the fine-tuned model to unseen texts.
5. Compare model predictions against the human gold standard.

NB! Bare in mind the Colab token/runtime limits!

Final remarks

1. Data collection and preprocessing claims up to 45-80% of the research time.¹
2. “trial_” → “main_student_groups” is not a plug-n-play!
3. Suggested research objectives for groups:

Group 1 : annotated dataset description + zero-shot evaluation

Group 2 : few-shots prompting evaluation

¹<https://www.anaconda.com/state-of-data-science-2020>

What is prepared for this teaching session

Data processing and analysis

Developed pipelines for:

- preprocessing raw Label Studio exports
- converting annotations into a analysis-ready dataset
- reporting annotation statistics (items, annotators)
- calculating inter-annotator agreement (IAA) and disagreements for heterogeneous annotation types
- identifying disagreements
- adjudication workflow

LLM experiments in CoLab

Developed and evaluated:

- zero-shot prompting workflows
- prompt engineering strategies
- structured JSON outputs
- tested alternative prompt formulations
- hierarchical annotation pipelines
- prompt debugging and error analysis
- computational feasibility assessments

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Project links

- Groups2-3 shared annotation project in *Deeplearning in NLP course* at Johannes's Label Studio workspace (reset and released to students on 15/06/2026, 23:57)
<https://app.heartex.com/projects?workspace=139506>
- github repository containing
 - raw output of Label Studio annotations
 - postprocessing script (transforming annotations to dataset, human-human IAA calculations, majority voting gold standard generation)
 - LLMs evaluation
 - preprocessed full data for Hansard 1980, 2025, 2026 (complete with speaker-party information)
 - a script to generate automatic annotations for Hansard data

<https://github.com/kunilovskaya/DLDH-SocSci>

Other definitions

`item == sent_id`

`label, tag, value ==` a choice made in the interface
(`no_group_mention`), or its internal code
(`AGE_CHILD`), or its formalized version
(`no_group_mention`, if selected $\rightarrow$
`group_mentioned`: “no”)

`incomplete` : `group_appealed` = yes but at least one of the
required appeal dimensions (reasoning, polarity, or
stance) was missing

`adjudicate` : annotator disagreement could not be resolved
automatically and requires manual review (e.g. triple
disagreement, a multi-label tag is not seen at least 3
times)

`disagreement` : disagreement score per item, the number of
annotation variables for which at least two annotators
provided different values. The higher, the more
contested the item.

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1. Convert two hierarchical negatives into a formal 0/1 category

LS output		dataset	
group_appeal_detection	stance	group_mentioned	group_appealed
no group mentioned	–	no	no
NaN	endorse	yes	yes
no group appealed	–	yes	no

2. Disagreement score and auto-adjudication²

binary	multiclass	multi-label
AAA → +0, adj=no	AAA → +0, adj=no	[A][A][B] → +1, adj=no
AAB → +1, adj=no	AAB → +1, adj=no	[A][B][C] → +2, adj=yes
ABB → +1, adj=no	ABC → +2, adj=yes	[A,B,C][A][D] +3, adj=no

²For reasoning, polarity, stance, and group_tags, disagreement is only evaluated when gold[“group_appealed”] = yes. Range: [0,8]; mean: 2.92

Trivia

1. keywords success rate: $154 / 183 \approx 84.2\%$.
2. annotatable content ratio (mentions/appeals): $93 / 154 \approx 60\%$
3. appeals to groups constructed as intersectional (to all appeals): $55 / 93 \approx 59.1\%$
4. items requiring follow-up annotation: 8
5. items with formal annotation quality concerns: 5

References I

- [1] Artstein, R. & Poesio, M. (2008). Survey article: Inter-coder agreement for computational linguistics. Computational linguistics, 34(4), 555–596.
- [2] Google (2026). Gemma 4 Model Card.
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- [5] Qwen Team (2026). Qwen3.6-27B: Flagship-level coding in a 27B dense model.
- [6] Team, Q. (2025). Qwen3 technical report.